

Sobriety Suite - Datasheet and User Manual

Portable Impairment Detection Device

General Information

Features

- Blood alcohol level, heart rate and reaction time testing
- WiFi functionality for immediate display and storage of results
- Compact, light and portable packaging for field test applications
- Noise minimization through dual powering circuit
- Rechargeable main battery
- Breathalyzer sensor with high sensitivity to alcohol and low sensitivity to benzine [cite mq3]
- Heart rate monitor with custom mount for measurement consistency
- Adjustable volume for audio cues supporting the tests

General Description

The Sobriety Suite is an embedded device designed to assess potential driver impairment through three independent physiological tests: a breathalyzer measuring blood alcohol content, a reaction time test, and a heart rate measurement. Results are displayed on an OLED screen mounted on the device and simultaneously served to a connected mobile device through a locally hosted web interface. This document covers the hardware architecture, firmware implementation, system characteristics and user instructions.

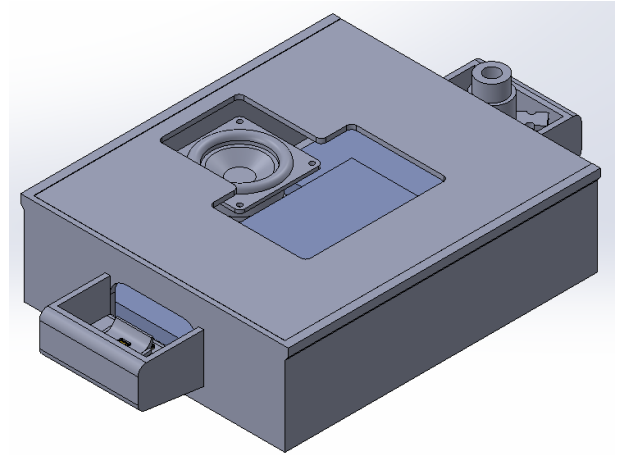


Figure 1: Sobriety Suite Revision 2 CAD

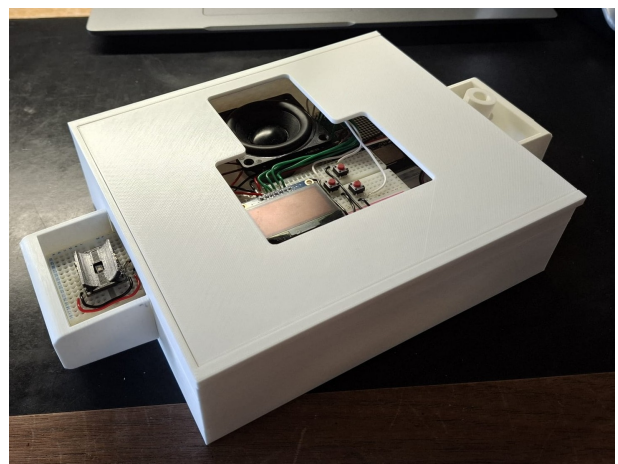


Figure 2: Sobriety Suite Revision 2 Final Product

Applications

- Student social events
- On campus bars, concerts, festivals and other events with limited medical personnel
- Assisting security personnel in student accommodation

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Electrical Specifications

Table 1: Specifications of the electronic components

Parameter	Symbol	Min.	Typ.	Max.	Unit	Comments
Speaker supply voltage (battery)	$V_{cc, speaker}$	9	9	12	V	range is within the operating range of the LM386 amplifier and avoids clipping [1]
Maximum main supply voltage	V_{pb}		5		V	”main supply” is the power bank connected through USB to the Arduino
Maximum main supply current	I_{pb}		1		A	
Main supply capacity	C_{pb}		2600		mAh	
Warm up time	T_0	500	600		s	based on alcohol sensor 0 reading calibration curve
Alcohol sensor temperature rating	t_{mq3}	-10	20±2	50	°C	[2]
Heart rate sensor temperature rating	t_{hr}	-40	25	85	°C	[3]

Mechanical Specifications

Table 2: Specifications related to the packaging

Parameter	Value
Overall system weight without power bank	539±1g
Maximum height	5.6±0.1cm
Maximum length	27.0±0.1cm
Maximum width	16.2±0.1cm

Circuit Schematic

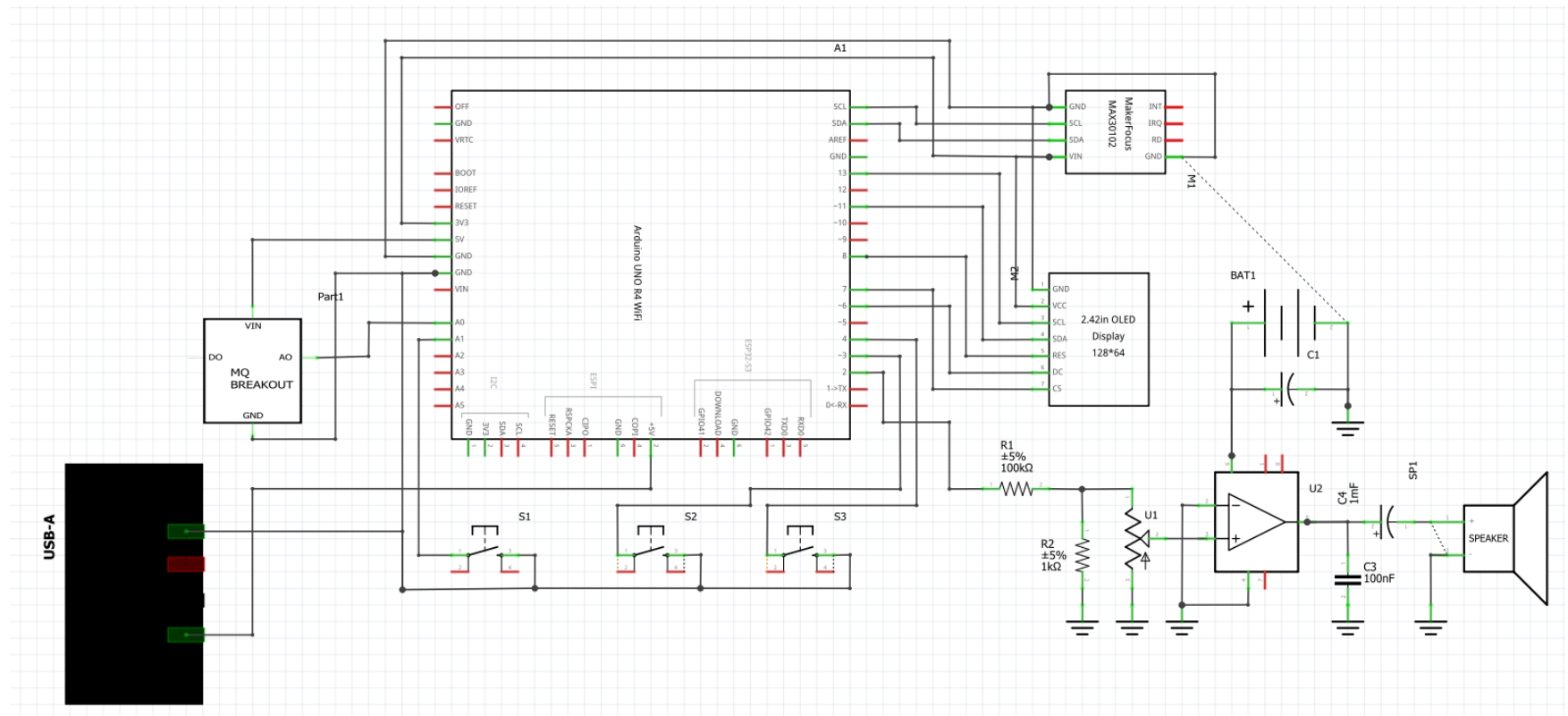
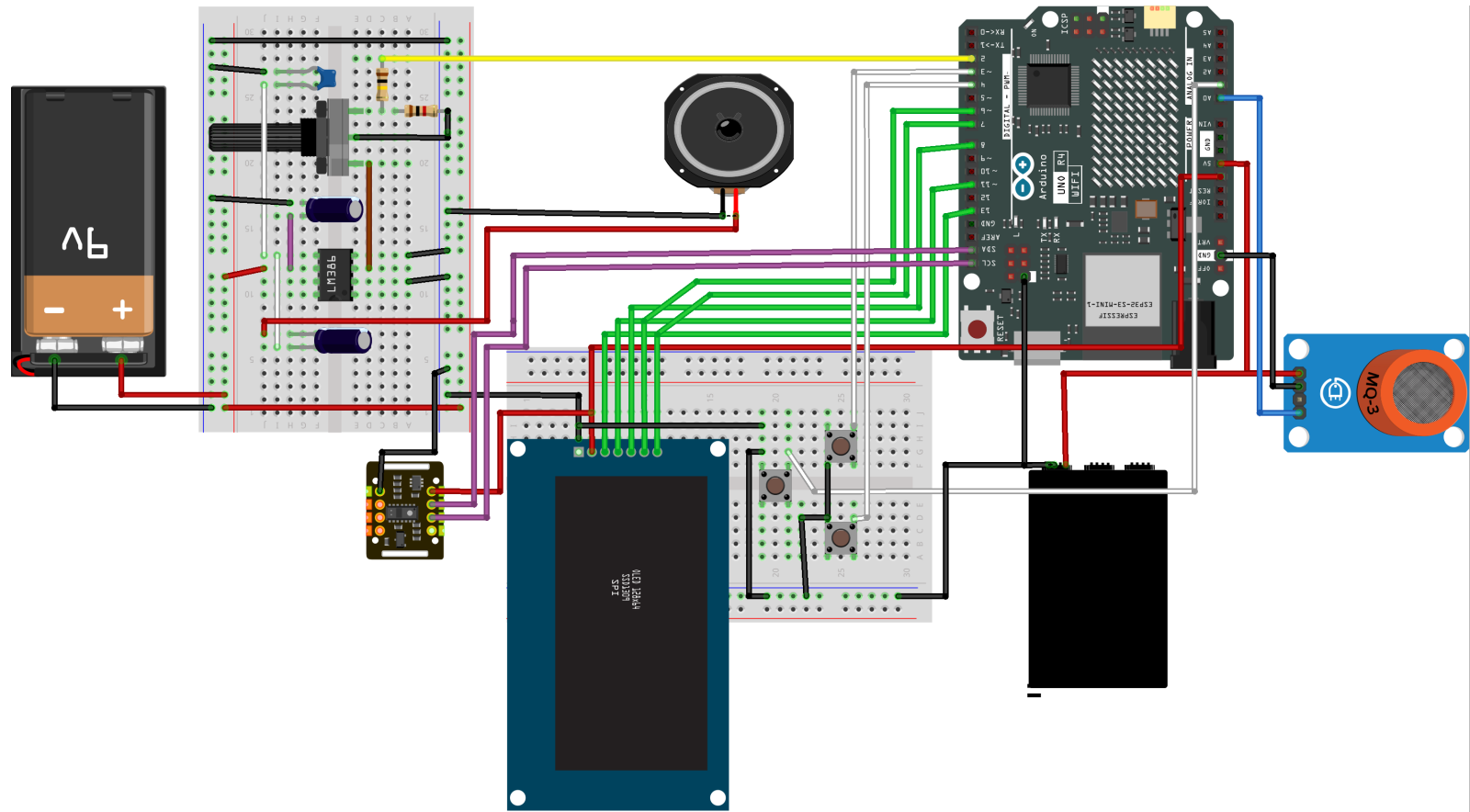


Figure 3: Circuit schematic including all Pin-outs

Wiring Diagram



fritzing

Figure 4: Wiring Diagram including all Peripherals and Powering Circuits

System Overview

System Block Diagram

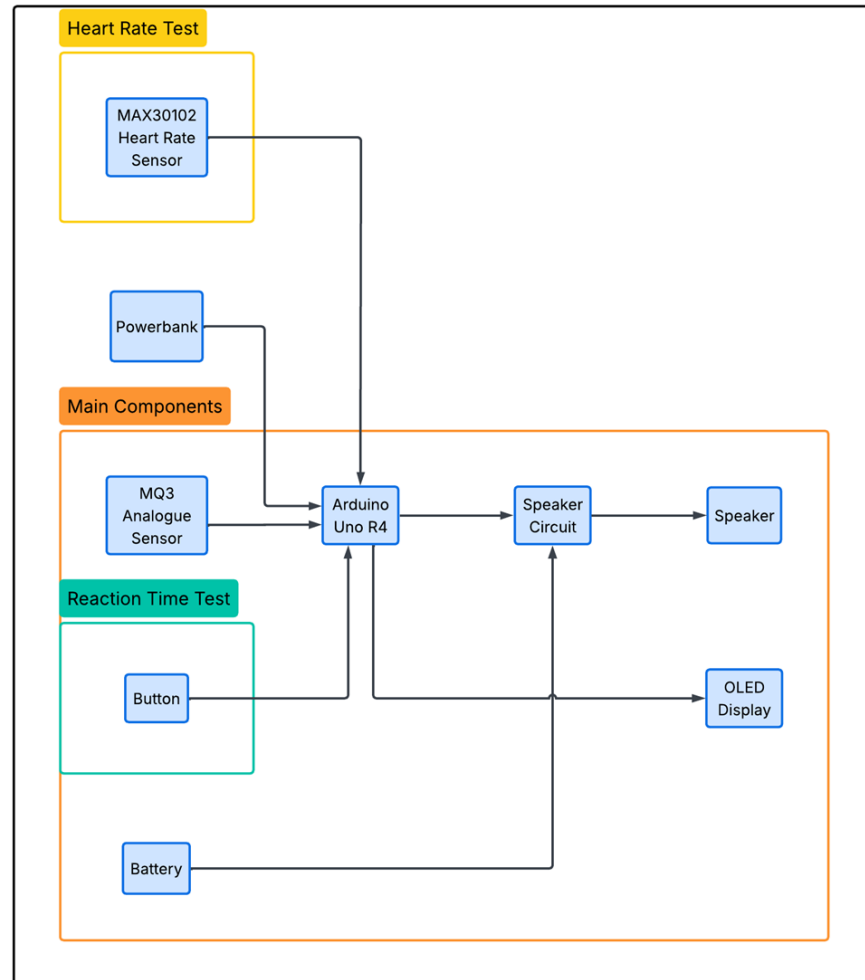


Figure 5: Block Diagram of the Sobriety Suite System Revision 2

Detailed Description of Blocks

Arduino Uno R4 WiFi

The R4 WiFi uses the Renesas RA4M1 microcontroller and includes an ESP32-S3 co-processor handling WiFi, exposed through the WiFiS3 library.

The R4 WiFi provides a stable 5V output pin sourced from USB independently of its 3.3V regulated rail, meaning both voltage levels are available simultaneously without contention. This eliminates any switching noise caused by regulators and provides a more stable supply to the MQ3 sensor in comparison to revision 1, which will be discussed in the "changes from the last revision" section.

Dual Powering Design

The power architecture splits the supply across two sources. A USB power bank is used to supply the Arduino R4 through its USB-C input, powering the microcontroller, display, sensors, and buttons. A separate 9V battery is used solely to drive the speaker through the LM386 audio amplifier. This isolation of the two power supplies leads to an extended battery life and the use of a rechargeable power bank also introduces a more sustainable usage model compared to replacing disposable batteries.

OLED Display - the Main User Interface

The display used is the MDOB128064BV-WS, a 128×64 pixel monochrome OLED panel driven by an SSD1306 controller, interfaced using the Adafruit SSD1306 library.

The SPI hardware pins of the Arduino are used for communication with the display and on a software level it is interfaced with as a Adafruit SSD1306 object.

An additional manual reset sequence is performed in software during setup, toggling the reset pin HIGH, LOW, then HIGH with short delays to ensure correct and reliable initialization.

The display provides 64 pixels of vertical space. For the menu overview, size 1 is sufficient but size 2 is used for prominent result labels such as PASS, FAIL, and WARN. Text size 3 is only used for the "GO" prompt during the reaction test to achieve good visibility.

Menu System and Navigation via Buttons

Navigation is handled through three buttons: UP, DOWN, and SELECT, all configured with internal pull-up resistors such that they read HIGH at rest and LOW when pressed.

Button presses are detected using a debounced buttonPressed() function. When a LOW state is detected on the target pin, a 30 millisecond delay is introduced and the pin is sampled again to confirm the press is genuine rather than electrical noise. The function then blocks until the button is released before returning true, preventing a single press from being counted multiple times.

Five menu items are displayed: Breathalyser, Reaction, Heart Rate, Results, and Show IP, and are rendered as a scrollable list with each of the three test items displaying a small [OK] or [X] badge on the right side of its row once that test has been completed. A compact status bar at the top of the menu shows single-character Y, N, or dash indicators for each test result at a glance.

Responsiveness of the web serves is consistently monitored to ensure a stable connection and immediate transfer of results.

Breathalyzer Sensor

The MQ3 is a chemiresistive gas sensor whose internal resistance varies in response to alcohol vapour concentration. The Arduino reads this as an analogue voltage on pin A0 through a voltage divider on the sensor module, converted to a 10-bit ADC value in the range 0 to 1023.

The sensor requires a significant preheat period before producing stable readings. This is implemented as a 10-minute warm-up countdown displayed on boot, with a progress bar and MM:SS timer. The user may skip this with SELECT if the sensor has already been warmed up. During the warm-up, raw ADC values are printed to the Serial monitor in comma-separated timestamp and value pairs, intended for logging to a spreadsheet for calibration analysis.

A 3D printed adaptor is provided along with disposable mouth pieces to ensure not only hygiene, but also a consistent distance from the sensor during the test to increase accuracy of the results.

The user is lead through the test using a series of visual and audio cues aimed at standardizing test duration. This further improves the consistency of the sensor behavior and accounts for the delay between change fo alcohol content in the air and change of internal resistance.

Reaction Time Test

The reaction test measures the time elapsed between a visual stimulus and a button press, used as a proxy for cognitive processing speed. After a preparation screen, the test introduces a random delay between 1000 and 3000 milliseconds before displaying the GO prompt in large text, preventing the user from anticipating the stimulus timing.

If no press is detected within a time window of 3000 milliseconds, the reaction test is automatically terminated and recorded as a fail.

The pass threshold is 500 milliseconds which is the sum of the average human reaction time (300ms), system delays and an allowable deviation from the average. Because the Arduino executes instructions sequentially and the button detection function introduces a 30 millisecond software debounce delay, a small but non-trivial latency is inherently present between the physical button press and the moment it is registered by the firmware. The test is therefore best interpreted as a relative indicator of impairment rather than a precise psychometric measurement.

Heart Rate Sensor

The heart rate sensor is the DFRobot SEN0344 module, built around the Maxim MAX30102 chip. The MAX30102 uses photoplethysmography, emitting infrared and red LEDs into the fingertip and measuring the reflected light intensity, which varies with the pulse of blood through the capillaries. The onboard MCU on the DFRobot module processes this raw optical data and outputs heart rate and SpO2 values over I2C at address 0x57.

The MAX30102 produces highly variable readings depending on finger placement, applied pressure, ambient temperature, and movement. A 3D printed jig is included based on consultation with the 3C11 module lecturer, Dr. Friedrich Wetterling. It standardizes finger positioning using the index finger placed between the distal and intermediate phalange over the sensor window.

To reduce the effect of momentary noise and outlier readings, the test samples continuously for 30 seconds and computes a running average of all readings that fall within the physiologically plausible range of 30 to 250 BPM.

The pass threshold is 130 BPM average. Elevated resting heart rate can be associated with alcohol intoxication as well as other physiological stressors. As with all components of this device, the heart rate measurement is presented as a supplementary indicator rather than a definitive clinical assessment, and the device explicitly makes no claim to medical accuracy or certification.

Web Interface

The Arduino UNO R4 WiFi operates as a WiFi access point, broadcasting the SSID SobrietySuite at the fixed IP address 192.168.4.1. A WiFiServer instance listens on port 80 and serves two endpoints: the root path returns the full HTML page, and the /data path returns a JSON object containing the current state of all test results.

The page is served directly from flash memory as a sequence of `client.println()` calls and requires no external resources, meaning it loads fully from the device alone with no internet connection required. Each test is presented as a card row with a colour-coded badge indicating pending, pass, warning, or fail state. The JavaScript on the page polls the /data endpoint every three seconds and updates all badge states and verdict text in place without a page reload, ensuring results appear on the connected device shortly after each test completes on the hardware.

The `handleWebClient()` function is called from every major loop in the firmware including the menu, the warm-up countdown, and the result waiting screens, ensuring HTTP requests are serviced continuously despite the single-threaded execution model of the Arduino runtime.

Audio Feedback

The speaker is driven by a digital square wave output from the Arduino to produce three distinct sound sequences for varying audio cue purposes. The 5V amplitude of this input signal is first scaled down using a voltage divider and then scaled up using a potentiometer and LM386 amplifier circuit to provide a volume adjustment option. Coupling and decoupling capacitors as well as a separate power supply ensure minimum noise at all volume levels.

Updates since Revision 1

Revision 2 includes significant improvements in term of resistance to noise due to the introduction of a dual powering system and a change from the Arduino Nano 33 IoT to the Uno R4 WiFi. This allowed the MQ3 alcohol sensor to be powered directly by the Arduino without the need for a buck converter, which had previously contributed to the noisy ground rail.

The background "hum" which occurred in the speaker in revision 1 has been successfully removed by implementing these changes. There is little to no audible noise over the entire volume range.

The MQ3 sensor warms up sufficiently within 10 min due to the larger supply current instead of the 30 to 45min in revision 1. The baseline output value has become more stable as well.

Additionally, the new powering system has a significantly longer battery life and allows for recharging of the main power supply, making it more sustainable.

Due to the change of microcontroller, an IMU is not available anymore to conduct a balance test. However, in the previous revision this test was found to be the least reliable and least meaningful, which is why no external tilt sensor was implemented in this version, opting to leave out the test entirely.

Characteristics and Sensor Behavior

MQ3 Sensor Warm-Up

The necessary warm-up period for the MQ3 sensor was determined to be 500 to 600 seconds based on experimental observations shown in Figure 6.

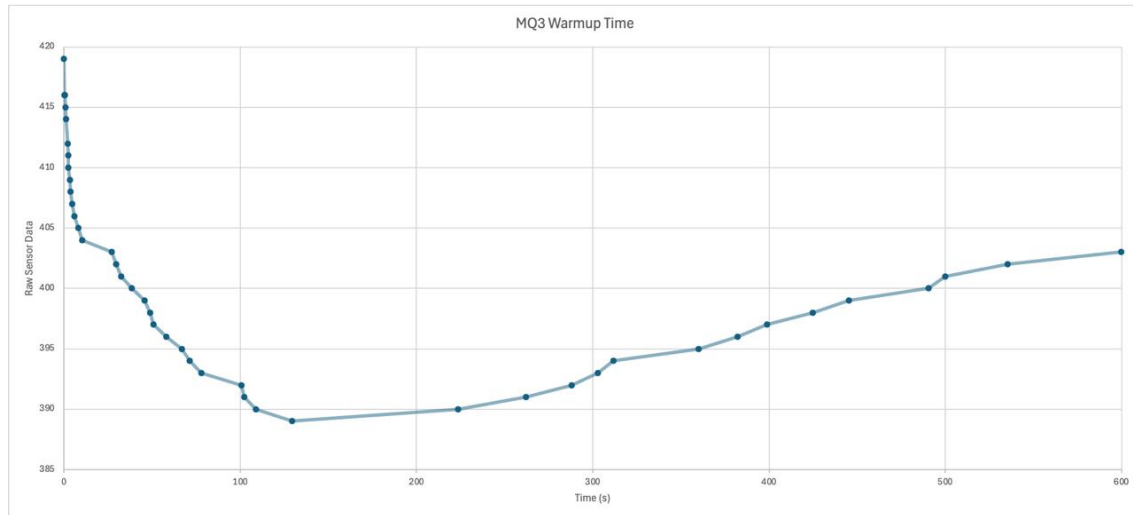


Figure 6: Baseline MQ3 Value in Air during Warm-Up Period

MQ3 Sensor Calibration

A sensitivity characteristic for the MQ3 sensor is given in its data sheet [2].

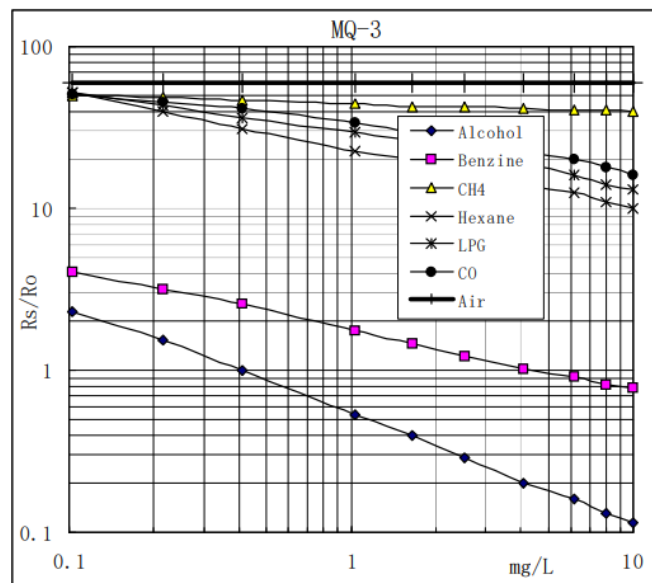
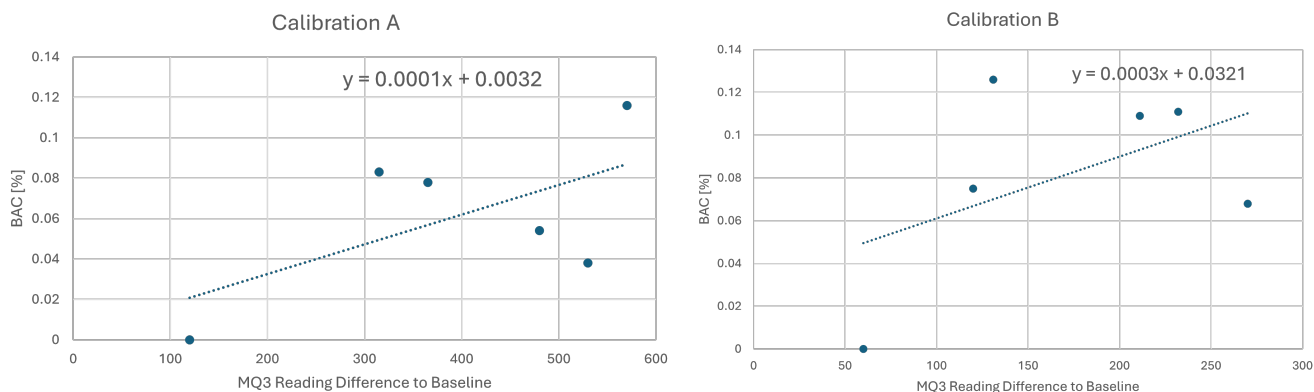


Fig.2 sensitivity characteristics of the MQ-3

Figure 7: Sensitivity Plot from the MQ3 Datasheet [2]

However, the digital output of the MQ3 is already processed and there is no public calibration curve for this. However, the plot in Figure 7 was included to represent the influence of different gasses compared to alcohol, which shows the most significant reaction.

Thus, calibration curves were created manually, but they were finalized before the changes to the powering circuit and thus represent the less accurate performance of revision 1. Revision 2 calibration curves will be released in the future.



(a) Calibration A

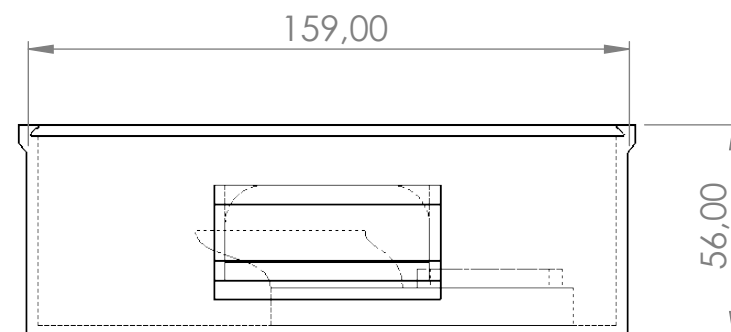
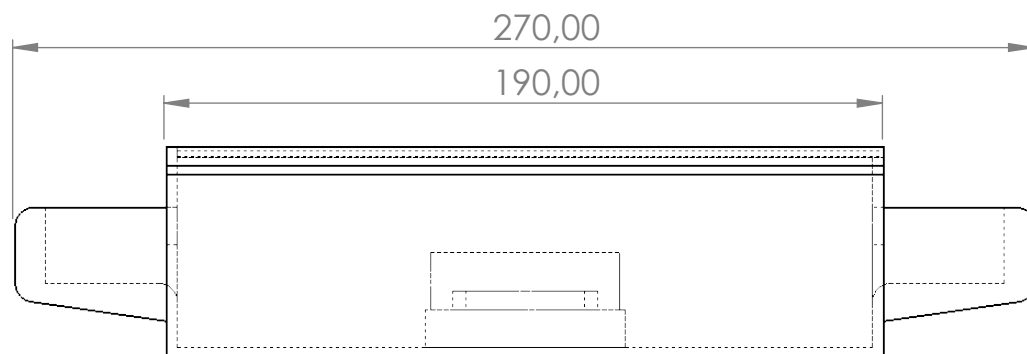
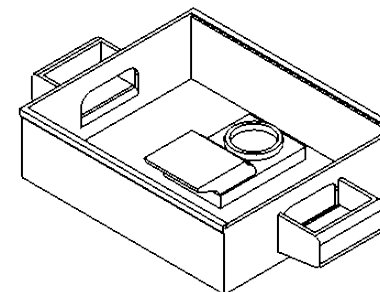
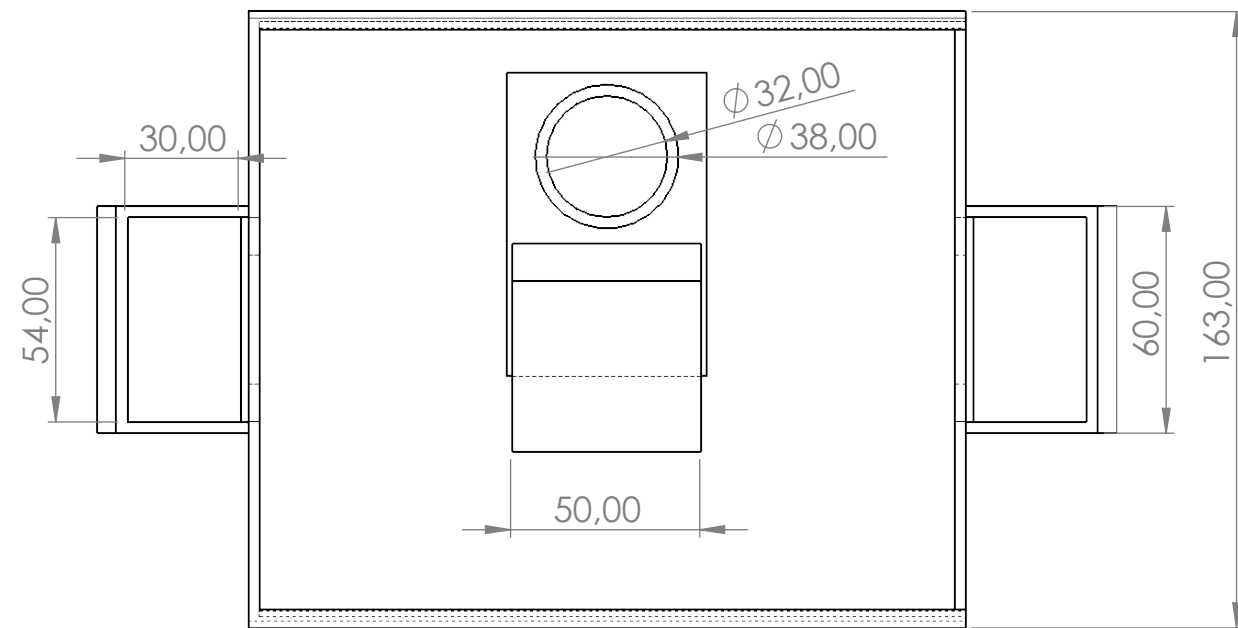
(b) Calibration B

Figure 8: Calibration Curves from Revision 1 (including Noise and Standardization Issues of Revision 1)

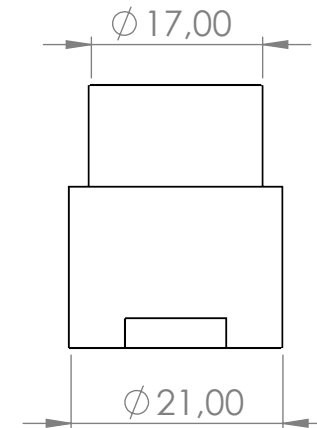
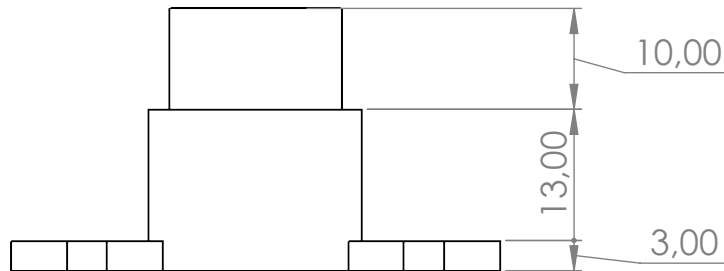
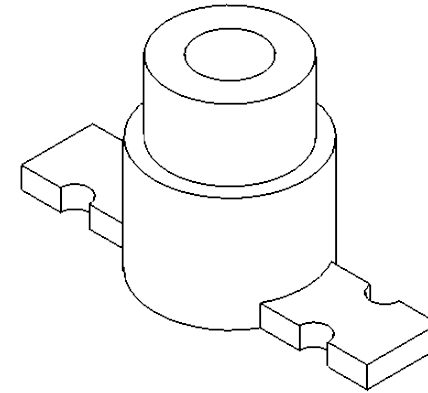
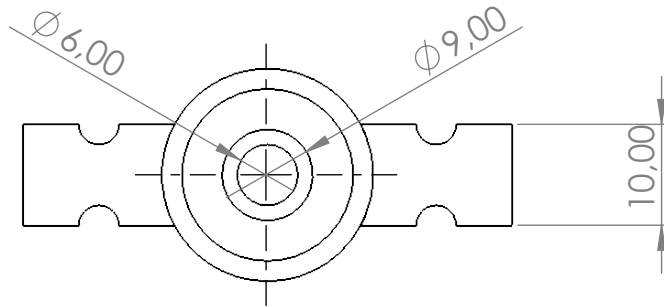
The final calibration equation is thus

$$BAC[\%] = \begin{cases} 0.0002 * \Delta MQ3, & \text{if } \Delta MQ3 \text{ is larger than } 120 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

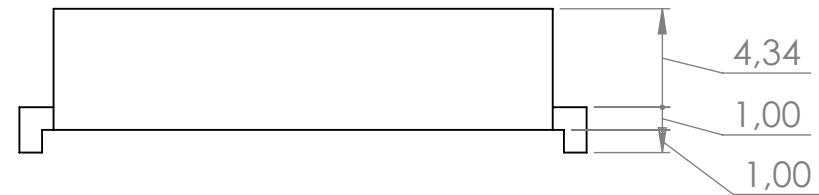
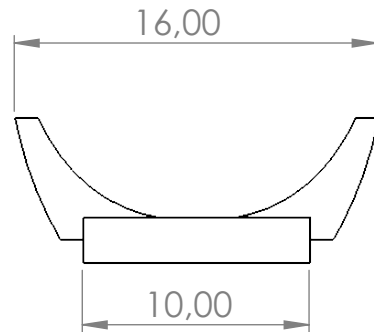
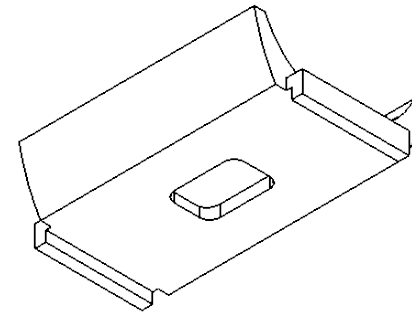
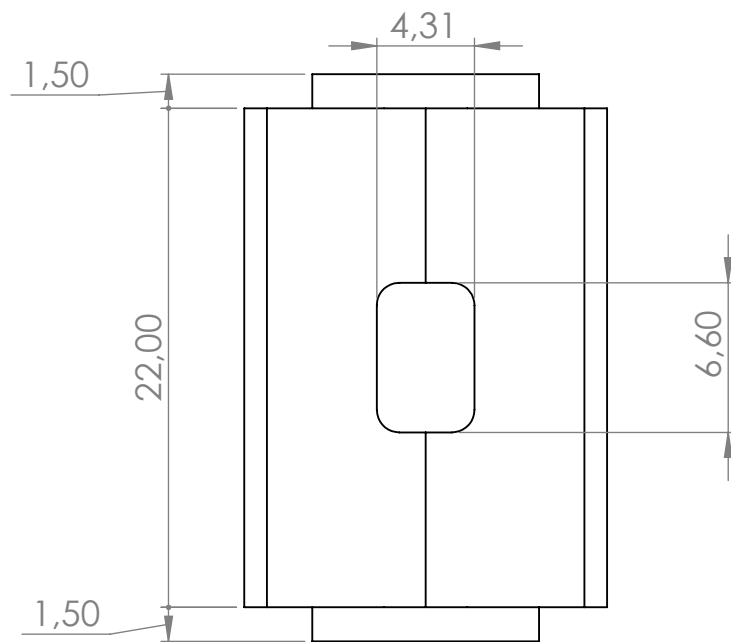
Technical Drawings



Drawing Properties	
Part	Sobriety Suite Main Caseing
Revision	2
Month	04/2026
Drawing Type	minimum dimensions for public data sheet
material	PLA



Drawing Properties	
Part	Sobriety Suite Breathalyzer Adaptor
Revision	2
Month	04/2026
Drawing Type	minimum dimensions for public data sheet
material	PLA



Drawing Properties	
Part	Sobriety Suite Heart Rate Sensor Adaptor
Revision	2
Month	04/2026
Drawing Type	minimum dimensions for public data sheet
material	PLA

User Manual

Important Disclaimer:

The Sobriety suite is a prototype educational device and is **not a certified medical device**. The results shouldn't be used as the sole basis for any safety critical decision. Always exercise personal judgment and comply with the applicable laws.

1. Getting Started

Power On

Connect the USB powerbank to the Arduino UNO R4 WiFi via the USB-C port. Switch the power bank on. The OLED display will illuminate and the device will begin a 10 minute sensor warm up countdown automatically. The MQ3 sensor requires approximately 10 minutes of preheat before it produces stable alcohol readings. A minutes and seconds countdown and a progress bar are shown on the display. Press SELECT to skip only if the device has been powered on and running for at least 10 minutes in the present session, skipping prematurely will result in unreliable breathalyzer readings.

Tip:

Refer to section XXX first, the web interface can be opened before any tests are complete. Results appear in real time as each test finishes on the device. If updates stop, double check your phone has not silently switched away from the SobrietySuite network as some devices drop WiFi connections that have no internet access.

Button Controls

- UP - scroll up using the top button
- DOWN - scroll down using the bottom button
- SELECT - confirm/proceed using the middle button

Main Menu

After the warmup, the main menu is displayed. 5 options available:

1) Breathalyzer	Alcohol breath test
2) Reaction	Reaction time test
3) Heart Rate	Resting heart rate measurement
4) Results	View combined test summary
5) Show IP	Display the device's web interface address

Navigate UP/DOWN and press SELECT to enter a test. Completed tests show a OK or X badge beside it. A compact status bar at the top of the menu shows a single character summary for all three tests at a glance.

2. Running the Tests

2.1 Breathalyzer

1. Select Breathalyzer from the menu and read the instruction screen. Press SELECT when ready.
2. A 3,2,1 countdown is displayed. Prepare to blow steadily.
3. When GO appears, blow steadily into the nozzle for approximately 2 seconds. Keep your mouth approximately 2-3 cm from the nozzle tip.

4. A 30 second settling period follows with an animated countdown and a melody. Do not blow again during this time.
5. Your estimated BAC and result is displayed.

Result	BAC range	Meaning
PASS Sober	< 0.02%	Within safe range
WARN Borderline	0.02% - 0.05%	Approaching legal limit
FAIL Intoxicated	> 0.05%	Above EU legal driving limit.

Hygiene: The sensor jig uses replaceable nozzle tips, Replace the nozzle tips with each user.

2.2 Reaction Test

1. Select Reaction from the menu, Read the preparation screen and then press SELECT.
2. A random delay of 1 to 3 seconds is introduced before the stimulus appears, Don't hold SELECT during this time.
3. When GO appears in large text, press SELECT button as quickly as possible.
4. Your reaction time in millisecond and pass/fail results are displayed.

Result	Condition
PASS	Response time within 500ms
WARN	Response between 500ms and 3000ms
FAIL	Not pressing the button within 3 seconds

Note on Accuracy: Due to software based button polling and a 30ms debounce delay, recorded times are slightly longer than the actual true physiological reaction time. The test is best interpreted as a relative indicator of impairment rather than an instrument that provides precise measurement.

2.3 Heart Rate Test

1. Select HEART RATE from the menu.
2. Place the tip of your index finger across the sensor window, refer to section 8 for correct placement.
3. Hold completely still for 30 seconds while the device samples and averages your heart rate.
4. Your average BPM and results are displayed.

Result	Condition
PASS	Average BPM $\leq$ 130
FAIL	Average BPM > 130, or not valid reading obtained

Note on Best Practice: For the most consistent reading, sit still, breathe normally and ensure ambient light is not directly shining onto the sensor window. Readings outside the 30-250BPM range are automatically discarded.

3. Viewing the Results

Select RESULTS option from the main menu to see a summary of all 3 tests with the respective individual PASS/FAIL states and an overall verdict of either SOBER or IMPAIRED. Pressing SELECT from this screen clears all stored results and returns the device to a clean state ready for another assessment.

4. Web Interface

1. Select Show IP from main menu. This displays the device's IP address, 192.168.4.1.
2. On your phone or laptop, open the WiFi settings and connect to the network named Sobriety Suite. Password is 'test12345'.
3. Open a browser and navigate to <http://192.168.4.1>.
4. The results page loads and updates automatically every 3 seconds as tests are completed, refreshing the page is not required.

5. Audio Feedback

The device uses distinct sound sequences to indicate test outcomes and device status.

- Four note ascending melody → Plays immediately after a PASS result on a test.
- Four note descending melody → Plays immediately after a FAIL or WARN result on a test.
- Looping eight note melody → Plays continuously during the 30 second breathalyzer settling period. This is normal, DO NOT blow again during playing melody period.

Speaker power: Audio powered by 9V battery connected to speaker circuit. If no sound heard, check if battery is sufficiently charged and connected. The battery doesn't affect any other function of the device.

6. Overall Verdict

- **ALL CLEAR**: All 3 tests passed. No significant indicators of impairment detected.
- **CAUTION**: Exactly 1 test failed. Retest recommended before driving.
- **DO NOT DRIVE**: 2 or more tests failed. Multiple indicators suggest impairment.

The verdict only reflects tests that have been completed. For a complete and reliable assessment, all three tests should be run. While tests are in progress the web page displays an **In Progress** state showing how many of the 3 tests have been completed.

7. What to do after a Fail

Safety note: A fail results on a test indicating potential impairment. DO NOT operate a vehicle or machinery. The device does not determine fitness to drive. **Breathalyzer fail or warning:** Wait at least 30 to 60 minutes before retesting and avoid consuming any further alcohol. To retest, return to main menu and SELECT breathalyzer again. **Reaction time fail or warning:** Slow reaction can results from fatigue, alcohol or distraction. Rest for a period before retesting. If results remain consistently slow after adequate rest, this may indicate impairment beyond fatigue. **Heart rate fail:** An elevated resting heart rate can be caused by alcohol, physical exertion, stress or caffeine intake. If you have been physically active recently, rest for atleast 5 minutes before retesting. If reading remains elevated at rest, DO NOT drive.

8. Heart Rate Sensor Placement Guide

Correct finger placement is the single most important factor in obtaining a reliable reading. Poor placement is the most common cause of a 0 BPM reading or erratic values.

- Use index finger of either hand.
- Rest the finger pad just below the finger nail.
- Apply light, consistent pressure, enough to maintain contact.
- Keep finger completely still for the full 30 seconds.
- Ensure no direct ambient light is falling on the sensor window.

9. Troubleshooting

Symptom	Likely cause and fix
Display does not turn on	Check the USB power is switched on and USB-C cable fully seated. Try pressing power bank button once, if still blank press Arduino's onboard reset button.
Breathalyser reads 0.00% even after blowing	The sensor may not have fully warmed up. Allow for a 10 minute warm up to complete and retest.
Breathalyser reading seems unrealistically high at rest	The MQ3 sensor is sensitive to other volatile organic compounds such as hand sanitizer and solvents.
No audio/speaker is silent	Speaker powered by separate 9V battery, check if connected and charged.
Heart rate sensor shows 0 BPM or no reading	Reposition finger, sensor window must be in direct contact with skin. Ensure finger held still. If 0 persists press SELECT to exit and redo the test.
Cannot connect to SobrietySuite WiFi	Ensure device is powered on and past the warm up screen, access point only starts after boot completes.
Web page loads but results don't update	Check that phone hasn't switched away from the SobrietySuite network, some devices drop connections with no internet access. Reconnect.

References

- [1] alldatasheet.com, “LM386 Datasheet, PDF.” Accessed: Apr. 14, 2026. [Online]. Available: <https://www.ti.com/lit/ds/symlink/lm386.pdf>
- [2] “Gravity: Analog Alcohol Sensor (MQ3) - Precision & Ease for Arduino!,” Gravity: Analog Alcohol Sensor (MQ3) for Arduino Projects — DFRobot Wiki. Accessed: Apr. 14, 2026. [Online]. Available: <https://wiki.dfrobot.com/sen0128/>
- [3] “MAX30102–High-Sensitivity Pulse Oximeter and Heart-Rate Sensor for Wearable Health”. Available: <https://www.analog.com/media/en/technical-documentation/data-sheets/MAX30102.pdf>